

What is Playwright?

Playwright is an open-source end-to-end automation testing framework developed by Microsoft. It allows you to automate web applications across Chromium, Firefox, and WebKit using a single API.

Key Capabilities

- Cross-browser testing (Chromium, Firefox, WebKit)
- Supports JavaScript, TypeScript, Python, Java, .NET
- Auto-waiting for elements
- Powerful selectors
- Built-in test runner
- Parallel execution
- Screenshots, videos, and trace viewer
- Mobile emulation and geolocation mocking
- API testing support

Why Playwright for Automation Testing

Playwright is fast, reliable, and designed to reduce flaky tests. It automatically waits for elements, network calls, and page stability, making tests more stable compared to traditional tools.

How to Configure Playwright

1. Install Node.js
2. Create a project folder
3. Run: `npm init playwright@latest`
4. Choose TypeScript or JavaScript
5. Playwright installs browsers automatically
6. Configure `playwright.config.ts`

Basic Playwright Test Flow

1. Launch browser
2. Create browser context
3. Open page

4. Perform actions (click, type, etc.)
5. Add assertions
6. Close browser

Playwright Best Practices

- Use auto-waiting assertions
- Prefer locators over selectors
- Use storageState for login
- Run tests in parallel
- Enable trace viewer for debugging
- Avoid hard waits

Conclusion

Playwright is a modern automation framework suitable for UI, API, and end-to-end testing. It is widely used in enterprise-grade automation due to its speed, stability, and rich debugging features.